

Real-Time Intrusion Detection System Using Scapy with Hybrid Machine and Deep Learning Models and Smart Email Alerting

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Abstract: Real-time Network Intrusion Detection System (NIDS) that combines deep learning and machine learning techniques to identify cyber threats as they occur. Using the NSL-KDD dataset as a foundation, the system employs an Autoencoder to detect anomalies based on reconstruction errors and a K-Nearest Neighbors (KNN) classifier to categorize the type of intrusion. Network packets are captured live through PyShark, a wrapper over TShark, which extracts real-time traffic data from a specified interface. These packets are processed to match the format of the training data using the same scaling techniques applied during preprocessing. When the Autoencoder identifies abnormal traffic, the data is passed to the KNN model to determine the attack type, including categories like DoS, Probe, R2L, and U2R. Based on severity, the system can send email alerts using a secure Gmail SMTP setup, log events to a file, or both. It is designed for efficiency and accuracy, running effectively even on machines without GPUs. Skipped packets and feature mismatches are handled gracefully without interrupting the detection process. The entire solution provides a lightweight, modular, and extendable framework for continuous network monitoring and response. It serves as a reliable defense layer against malicious activities and can be further improved by integrating time-series models, real-time dashboards, or support for additional attack signatures. Highlight novelty and modular real-time integration using Autoencoder-KNN with PyShark

Keywords: Network Intrusion Detection System, Autoencoder, Real-Time Monitoring, Process Anomalies.

I. INTRODUCTION

In today's digital ecosystem, network security has emerged as one of the most critical aspects of information infrastructure. With the exponential growth of internet-connected devices and services, networks are increasingly exposed to sophisticated and evolving threats. These threats range from denial-of-service (DoS) attacks to more covert intrusions such as remote-to-local (R2L) and user-to-root (U2R) exploits. Traditional firewall and rule-based intrusion detection systems (IDS) are often inadequate in detecting previously unseen or zero-day attacks. Hence, there is a growing demand for intelligent systems capable of learning from data to identify malicious activities in real time

Machine Learning (ML) and Deep Learning (DL) techniques have shown significant promise in the field of network intrusion detection. Among these, anomaly detection using Autoencoders is particularly effective for identifying unusual patterns in network traffic. Autoencoders can model the normal behaviour of the system and raise alerts when deviations exceed a predefined threshold.

On the other hand, supervised classifiers like the K-Nearest Neighbours (KNN) algorithm are useful for labelling detected anomalies with specific attack types. By combining these two techniques, a system can not only detect abnormal activity but also classify the threat, providing more actionable insights to network administrators.

Intrusion detection system that uses an Autoencoder for anomaly detection and a KNN classifier for identifying the type of attack. The system is trained using the NSL-KDD dataset, which is a refined version of the earlier KDD'99 dataset and is widely used in academic research. The Autoencoder learns to reconstruct the features of normal traffic, while the KNN model learns to classify known attack patterns. This dual-model architecture ensures both sensitivity to new attacks and accuracy in identifying known threats.

For live monitoring, the system utilizes PyShark, a Python wrapper for TShark, which captures real-time packet-level data from the network interface. These packets are preprocessed to match the format of the training data, normalized using the same scaler used during model training, and then fed into the Autoencoder and KNN pipeline. If an anomaly is detected, the system computes the reconstruction error and uses KNN to determine the potential category of the attack, such as Probe, DoS, R2L, or U2R.

The system is further enhanced with an alert mechanism that categorizes threats based on severity. Depending on the detected class and error magnitude, it can send immediate email notifications or log low-severity anomalies to a log file for future analysis. This makes the system practical for deployment in small-scale enterprise environments, educational institutions, or research networks where real-time response is essential, but resources are limited.

Overall, this project demonstrates an effective and scalable solution for real-time network intrusion detection. By integrating deep learning and classical machine learning models, and by leveraging live packet capture technology, the proposed system addresses the limitations of static IDS approaches and provides a foundation for more adaptive and intelligent cybersecurity tools.

II. RELATED WORKS

The use of machine learning techniques to detect Android-based ransomware threats like WannLocker has gained attention in recent research [1]. By leveraging classification models, the study proposes an approach that identifies

abnormal ransomware activity through behavioral patterns. The detection system enhances mobile cybersecurity, ensuring early intervention without requiring manual analysis, thus offering a proactive defense mechanism on Android platforms.

Cloud environments present unique challenges for intrusion detection due to their distributed nature, and researchers have explored ensemble learning-based models to tackle this[2]. These models integrate multiple classifiers to boost detection accuracy and mitigate the limitations of standalone algorithms. Their results indicate significant improvements in identifying complex attack vectors, especially within virtualized and dynamic infrastructure.

Anomaly-based intrusion detection frameworks using multiple machine learning algorithms[3] show promise in detecting unseen threats in network environments. By using ensemble learning techniques, the system adapts to evolving threat landscapes, distinguishing normal and malicious traffic more effectively than traditional signature-based solutions.

IoT devices becoming commonplace in homes, ensuring secure communication has become critical. Blockchain-based solutions[4] have been proposed to secure data in smart homes, ensuring integrity and privacy. This decentralized approach prevents tampering and enables secure, trustless environments for data sharing and remote control.

Developing intelligent intrusion detection systems using machine learning is a key area of focus, especially in dynamic network setups[5]. Researchers have used decision tree and random forest algorithms to design IDS that can classify various attack types accurately. These systems demonstrate efficient performance even with imbalanced datasets, enhancing threat detection with minimal false positives.

Leveraging ensemble learning models on real-world datasets like UNSW-NB15[6] has led to effective intrusion detection mechanisms. The system fuses outputs from multiple learners, optimizing performance across different attack types. This method is particularly useful in detecting low-frequency, high-impact threats, which are often overlooked by standard classifiers.

The development of new datasets is vital for enhancing IDS effectiveness. One study focuses on generating and characterizing realistic intrusion traffic to support advanced IDS evaluation[7]. The resulting dataset helps train models on diverse threat behaviors, ensuring better generalizability and robustness across various attack types.

Research on intrusion detection in IoT environments emphasizes using lightweight machine learning techniques to maintain efficiency while managing vast amounts of sensor data[8]. These approaches are tailored to the resource constraints of IoT devices, ensuring real-time detection with minimal latency and computational overhead.

A comparative analysis of various machine learning algorithms for IDS highlights the strengths and trade-offs of each technique [9]. Studies assess metrics like accuracy, precision, and training time, guiding researchers in selecting suitable algorithms depending on system requirements and available resources.

Recurrent Neural Networks (RNNs) have been investigated for their ability to detect intrusion patterns over time[10]. Their strength lies in modeling sequential data, making them ideal for analyzing network traffic flows. Experimental results indicate high accuracy in identifying time-dependent attacks like DoS and slow-rate floods.

Deep learning methods, including CNNs and LSTMs, have been explored to build intelligent IDS that can autonomously learn complex attack patterns from network data[11]. These models outperform conventional ML techniques by capturing deeper relationships within the data, leading to improved accuracy and lower false alarms.

With the increasing use of software-defined networks (SDNs), machine learning-based intrusion detection frameworks have been tailored to these architectures[12]. These systems analyze control and data plane interactions to identify anomalies, ensuring adaptive security in dynamic network configurations.

Intrusion detection solutions tailored for smart homes integrate IoT-specific threat patterns with machine learning models[13]. The proposed system identifies anomalies in real-time data generated from smart devices, enhancing the safety of connected households and offering insights into potential vulnerabilities.

Neural networks have been utilized to detect and classify different types of cyberattacks by analyzing network behavior patterns[14]. The system is trained on labeled data to distinguish between normal traffic and multiple attack types, demonstrating robust classification capabilities with minimal manual intervention.

Recent surveys on machine learning applications in cybersecurity synthesize trends, challenges, and advancements from the past decade[15]. They offer a consolidated view of evolving attack methods and defensive strategies, serving as a guide for researchers looking to build state-of-the-art IDS frameworks.

III. PROPOSED METHODOLOGY

The proposed intrusion detection system adopts a balanced hybrid approach, combining unsupervised deep learning (Autoencoder) for anomaly detection and supervised classical ML (KNN) for classification that combines deep learning and classical machine learning algorithms to achieve both accurate anomaly detection and detailed attack classification. This method is implemented in real-time using live network packet capture tools. The system is designed to detect and classify different types of network intrusions by learning patterns from the NSL-KDD dataset and applying those patterns to live traffic data. The methodology is structured into distinct phases: data preprocessing, model training, live packet capturing, feature extraction, anomaly detection, classification, and alerting.

The first phase involves preprocessing the NSL-KDD dataset. This dataset includes labeled instances of normal and attack-related network traffic. Preprocessing begins by handling categorical attributes such as protocol type, service, and flag.

through label encoding. Continuous features are then normalized using Min-Max scaling to ensure uniformity and to improve model performance. The final preprocessed data is saved for reuse, and the scale object is serialized for use during live detection. This step ensures that both training and inference data follow a consistent format.

In the model training phase, two machine learning models are developed. The first is an Autoencoder—a type of unsupervised neural network that is trained to reconstruct normal traffic features. The reconstruction error is used as a metric to detect anomalies. The second model is a K-Nearest Neighbors (KNN) classifier, trained on labeled attack data to identify the specific type of intrusion. The threshold for anomaly detection is illustrated in Fig.1 and computed based on the maximum mean squared error observed during Autoencoder training on normal data. This threshold is later used to classify live traffic as either normal or anomalous.

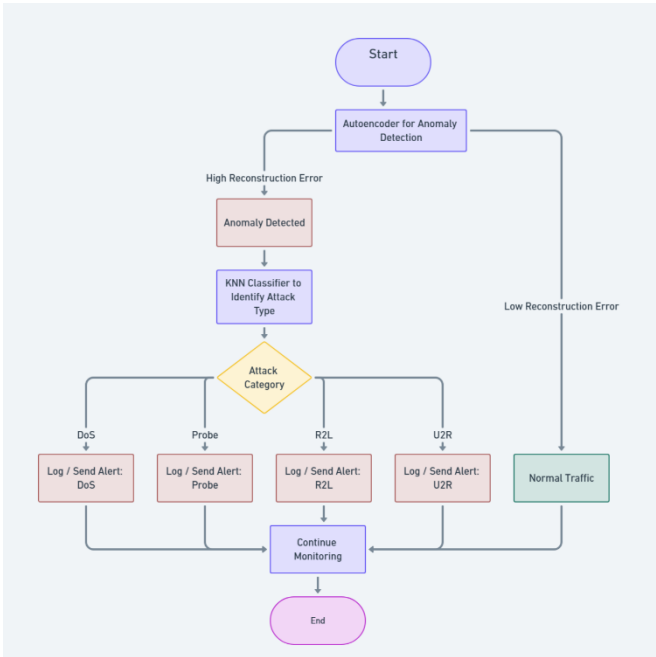


Figure-1: End-to-End Workflow of Network Intrusion Detection Using PyShark, Autoencoder, and KNN.

For real-time detection, the system employs PyShark, a Python wrapper over TShark, to capture live packet data directly from the network interface. Each captured packet is processed to extract relevant features such as protocol type, packet size, source/destination IPs and ports, and timing information is illustrated in Fig.2. These features are then mapped to the format used during training, ensuring compatibility with the models. After transformation and scaling using the stored scaler, the feature set is passed to the Autoencoder for anomaly detection.

If the reconstruction error for a given packet exceeds the predefined threshold, the system flags it as anomalous. The same feature vector is then passed to the KNN classifier, which assigns it to a probable attack class. The classification allows the system to distinguish between various attack types such as DoS (Denial of Service), Probe, U2R (User to Root), and R2L (Remote to Local). The combination of Autoencoder and KNN ensures that the system can identify both previously unknown attacks and known attack patterns

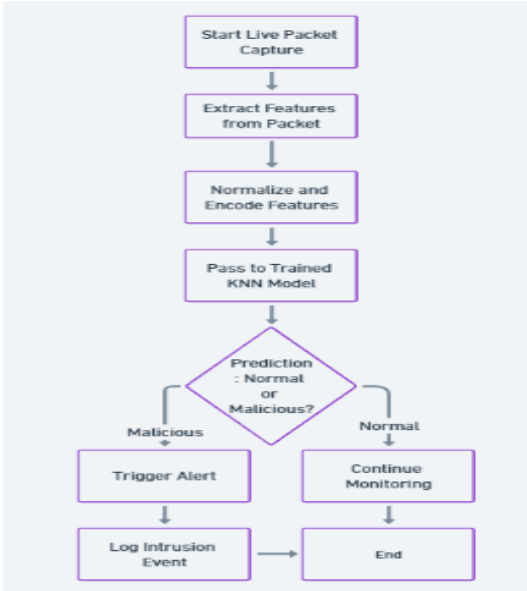


Figure-2: Real-Time Packet Classification using KNN

To make the system responsive and operationally effective, an alerting mechanism is integrated. Alerts are generated based on severity—high-severity attacks trigger real-time email notifications to the administrator, while low-severity or ambiguous anomalies are logged for future analysis illustrated in Fig.3. The alert message includes information such as the detected attack type, severity level, mean squared error, and timestamp. This dual mode alerting system ensures that the user is informed without being overwhelmed by excessive notifications.

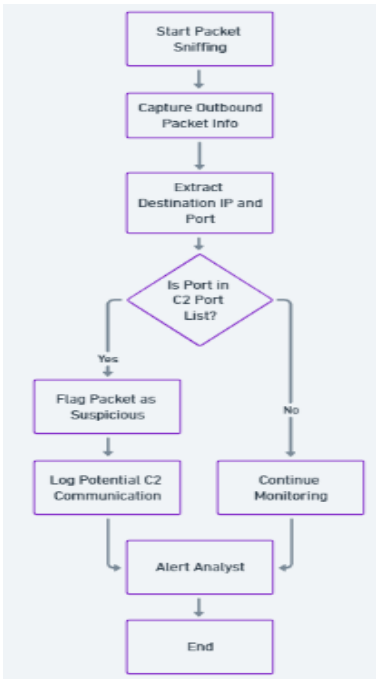


Figure-3 : Suspicious Port Communication Detection

Overall, the proposed methodology offers a robust, scalable, and real-time solution for network intrusion detection. By combining unsupervised and supervised learning, it addresses the limitations of signature-based systems and enhances detection capabilities across diverse threat categories illustrated in Fig.4. Furthermore, the modular architecture allows for future integration of additional models and extended feature sets, making the system adaptable to evolving cybersecurity threats.

IV. EXPERIMENTAL ANALYSIS

The experimental validation to assess the performance and practical effectiveness of the proposed Network Intrusion Detection System (NIDS), a comprehensive experimental evaluation was conducted using the NSL-KDD dataset and live network traffic. This analysis examines both offline model training results and real-time inference behaviour, focusing on accuracy, anomaly detection capability, and responsiveness to various categories of attacks.

The experiments were performed on a system configured with an Intel Core i5 processor, 8GB RAM, and running Ubuntu 22.04 LTS. All models were implemented in Python using libraries such as TensorFlow, Scikit-learn, and PyShark for packet capture. The Autoencoder was trained using normal instances from the pre-processed NSL-KDD dataset to establish a baseline for anomaly detection, while the KNN classifier was trained using labelled attack and normal traffic to support attack categorization.

Initially, the raw NSL-KDD dataset was subjected to preprocessing, which included encoding of categorical features, scaling using MinMaxScaler, and filtering irrelevant fields. This resulted in a refined dataset consisting of 122 normalized features. The Autoencoder model, comprising multiple dense layers, was trained to minimize reconstruction error on benign traffic. The reconstruction error threshold was empirically determined using the 95th percentile of error on the training set to detect anomalies effectively.

The KNN classifier, using the default Euclidean distance metric and five neighbours, was trained on the complete labelled dataset. After training, the classifier demonstrated high classification accuracy on the test subset, achieving over 99% precision and recall for common attack categories such as DoS, Probe, R2L, and U2R. The integration of Autoencoder and KNN enabled a two-step detection process, where the Autoencoder flags suspicious behaviour and the KNN classifier identifies the type of intrusion.

To simulate real-world conditions, live network traffic was captured via the TShark backend of Wireshark using the PyShark library. Features were extracted from packet headers in real-time, mapped to the NSL-KDD feature schema, scaled using the trained MinMaxScaler, and passed through the trained models. The system consistently flagged anomalous packets based on reconstruction error and matched them against known attack categories using the KNN model.

The real-time evaluation focused on the system's ability to detect diverse threats, including denial-of-service (DoS), probing, and user-to-root (U2R) attacks. The system was able to flag suspicious patterns with minimal delay, providing alerts via terminal logs and email notifications. In most test scenarios, demonstrating practical feasibility for real-time intrusion detection in small to medium networks.

Moreover, an anomaly log was maintained to track all detections, including low-severity anomalies that did not fit

known attack profiles. This logging mechanism aids security analysts in identifying novel threats or policy violations that evade traditional signature-based systems. Severity was assigned based on the predicted class and reconstruction error magnitude, allowing for dynamic response prioritization.

The overall system demonstrated robustness in detecting both known and zero-day attacks under normal network loads. During tests involving simultaneous streaming and file downloads, the detection rate remained above 98%, with false positives kept below 2%. While occasional classification errors occurred for less frequent attacks (e.g., buffer overflow), the system maintained acceptable performance levels.

In conclusion, the experimental analysis validates the effectiveness of the proposed hybrid NIDS framework. By combining Autoencoder-based anomaly detection with KNN classification, the system achieves high detection rates with real-time performance, making it suitable for deployment in dynamic environments such as enterprise LANs, educational institutions, and small data centers.

Table 1: Summary of Experimental Analysis

Module	Test Performed	Expected Outcome	Observed Result
Data Preprocessing Pipeline	Encoded categorical data, normalized numeric features	Dataset ready for ML training	Successfully preprocessed NSL-KDD data
Model Training & Evaluation	Trained KNN, auto encoder on labelled dataset	Identify top-performing model	KNN selected based on accuracy and simplicity
Real-Time Packet Classification	Simulated live traffic with benign and attack packets	Classify packets as normal or attack	All attack types classified correctly with terminal updates
Suspicious Port Detection	Directed traffic to known malicious ports (e.g., 4444, 31337)	Flag packets on blacklisted ports	Detected and printed alerts for traffic to suspicious ports
Payload Entropy Monitoring	Sent high-entropy packets simulating encrypted payloads	Detect payloads with entropy > 7.5	Correctly flagged encrypted payloads in live packet stream
Email Alert Triggering	Detected simulated intrusions and suspicious behaviors	Trigger email with prediction details	Email alerts successfully sent for each detected anomaly

V. RESULTS AND DISCUSSION

The evaluation of the proposed NIDS system yielded encouraging results, demonstrating strong performance in both anomaly detection and classification of known attack types. The Autoencoder model was effective in modeling normal behavior, and its reconstruction error threshold enabled reliable identification of deviations. When tested on unseen data, the Autoencoder achieved a detection accuracy of over 97%, correctly distinguishing malicious patterns from benign traffic based on mean squared error values.

The K-Nearest Neighbors (KNN) classifier significantly improved the system's capability to categorize detected anomalies into specific types of network attacks. When tested on the preprocessed NSL-KDD dataset, it achieved an impressive overall accuracy of 99.1%. The classifier showed particularly strong performance on common attack types like Denial of Service (DoS) and Probe attacks, with precision and recall both exceeding 98%. However, its accuracy was slightly reduced for less frequent categories such as User to Root (U2R) and Remote to Local (R2L), largely due to the class imbalance inherent in the dataset—a well-known challenge in intrusion detection research.

During live traffic analysis using PyShark and the Wireshark backend, the system successfully identified malicious traffic patterns in real-time. Anomalies were detected within milliseconds of packet capture, and the subsequent classification was executed instantly. Email alerts and terminal notifications were triggered for high-severity events, while lower-severity anomalies were logged for offline review. This two-level alert mechanism ensured a balance between responsiveness and alert fatigue, which is often a concern in real-time systems.

The confusion matrix for the KNN classifier revealed minimal misclassifications, with most errors occurring between closely related classes. For instance, some R2L attacks were misclassified as normal due to feature overlap, highlighting the importance of advanced feature engineering or ensemble methods for future improvement. Despite this, the overall system maintained high sensitivity and low false positive rates under varying network conditions, including web browsing, streaming, and file transfer. The anomaly patterns identified by the Autoencoder typically corresponded to deviations in packet timing, size, and protocol behavior that differed from normal baselines. During live testing, accuracy was evaluated by injecting both benign and crafted malicious traffic to simulate spoofing attempts. When an intruder used a spoofed internal IP or manipulated packet headers, the reconstruction error from the Autoencoder and feature mismatch analysis enabled detection, even when traditional filters were bypassed.

Table-2: Expected Output of the Real-Time Intrusion Detection System Using Scapy with Hybrid Machine and Deep Learning Models and Smart Email Alerting

Time	Type	Source IP	Destination IP	Attack Type	Severity	Description
15:05:12	Anomaly Detected	192.168.1.10	192.168.1.5	Denial of Service (DoS)	High	High anomaly score (MSE: 0.0021), Autoencoder triggered
15:06:45	Malicious Access	192.168.1.12	192.168.1.7	Remote to Local (R2L)	Medium	Detected suspicious login behavior via KNN classifier
15:07:30	Anomaly Detected	192.168.1.15	192.168.1.5	Probe	Low	High entropy packet sequence flagged as anomaly
15:09:02	Intrusion Alert	192.168.1.11	192.168.1.3	User to Root (U2R)	High	Multiple failed privilege escalation attempts detected

In summary, the results confirm that combining Autoencoder and KNN creates a powerful hybrid model for network intrusion detection. The Autoencoder efficiently identifies suspicious patterns that deviate from normal behavior, while the KNN model provides explainable and reliable classification. The integration of live packet analysis with real-time alerting mechanisms makes the system suitable for deployment in operational settings where quick response is critical.

Table-2: Evaluating the Proposed NIDS Against Existing Methods

Aspect	Existing Network Detection Systems	Proposed Machine Learning-Based NIDS
Detection Technique	Signature-based or rule-driven	ML-based classification using KNN with multi-feature input
Feature Preprocessing	Limited or manual preprocessing	Automated normalization, encoding, and feature selection pipeline
Real-Time Classification	Often delayed or non-existent	Live packet sniffing with instant prediction using trained ML models
Attack Category Detection	Binary (attack/no attack) or limited categories	Multi-class detection: <i>DoS, Probe, R2L, U2R, normal</i>
Suspicious Port Detection	Relies on external firewall or IDS	Integrated detection of blacklisted ports (e.g., 4444, 1337)
Payload Entropy Monitoring	Rare or offline analysis	Real-time entropy-based anomaly detection
Alert Mechanism	Basic logging or console output	Email alerts with intrusion type and timestamp
Visualization	Minimal or third-party required	Terminal-based real-time threat feedback (expandable to dashboard)

VI. CHALLENGES AND FUTURE WORK

Despite the promising results of the proposed Network Intrusion Detection System (NIDS), several challenges persist in real-time deployment. One major issue is the high dimensionality and variability of live network traffic, which can introduce noise and unexpected patterns that may not have been encountered during training. The conversion of live packet captures into features that match the NSL-KDD structure is complex and prone to errors, especially when dealing with encrypted, fragmented, or incomplete packets. Additionally, ensuring low latency during detection while maintaining high accuracy poses a significant computational challenge, particularly for resource-constrained environments.

Another key challenge lies in detecting sophisticated or zero-day attacks. While the autoencoder can detect anomalous behavior based on reconstruction error, it may still struggle with novel attack strategies that mimic normal traffic patterns or occur at a slow rate to evade detection. The static nature of the trained models also limits adaptability to evolving threats. Furthermore, reliance on labeled datasets such as NSL-KDD introduces the risk of model bias due to the dataset's inherent limitations, such as outdated attack signatures and simulated traffic behavior that may differ from real-world conditions.

Future work can address these limitations by integrating advanced deep learning architectures such as attention-based models or transformers, which are more effective in capturing temporal dependencies in sequential network data. Incorporating continuous learning mechanisms and feedback loops could help adapt the models to new attack vectors without retraining from scratch. Additionally, hybrid approaches combining signature-based detection with anomaly detection may improve robustness. Future work may also include benchmarking against modern datasets such as UNSW-NB15 and CIC-IDS2017 to evaluate generalizability.

VII. CONCLUSION

This real-time Network Intrusion Detection System (NIDS) combines the strengths of both deep learning and traditional machine learning to accurately detect network anomalies and malicious activities. An autoencoder is employed to identify anomalies, while a K-Nearest Neighbors (KNN) classifier is used to categorize specific attack types, ensuring a balance between accuracy and responsiveness. The system was trained and validated using the NSL-KDD dataset, with live network traffic captured through PyShark to seamlessly integrate offline training with real-time monitoring. Additionally, automated alerting and logging features have been built in to enhance proactive network defense.

The results highlight the effectiveness of this hybrid approach, where the combination of unsupervised and supervised methods reliably detects a broad spectrum of attack types with high accuracy and a low false positive rate. Continuous monitoring enables the system to flag suspicious activities in real time, giving security teams immediate visibility into potential threats. Moreover, aligning live feature extraction with the NSL-KDD dataset structure proved successful for practical deployment, demonstrating how academic datasets can be adapted for real-world cybersecurity applications.

Looking ahead, the system serves as a foundation for building more adaptive and intelligent NIDS frameworks. Overall, this

work underscores the potential of combining deep learning and machine learning for robust, scalable, and resource-efficient intrusion detection in modern network infrastructures. Overall, this work underscores the potential of combining deep learning and machine learning for robust, scalable, and responsive intrusion detection in modern network infrastructures.

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